

dapper: Data Augmentation for Private Posterior Estimation in R

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Abstract This paper serves as a reference and introduction on using the dapper R package. It is an MCMC sampling framework which targets the exact posterior distribution given privatized data. The goal of this package is to provide researchers a tool to perform valid Bayesian inference on data protected by differential privacy. A strength of this framework is that it can be applied in situations where the likelihood is analytically intractable.

1 Introduction

Differential privacy provides a rigorous framework for protecting confidential information from re-identification attacks by using random noise to obscure the connection between the individual and data (Dwork et al. 2006). Its development was spurred on by successful attacks on anonymized data sets containing sensitive personal information. Prior to differential privacy, anonymization schemes did not always have sound theoretical guarantees despite appearing adequate. In one influential incident, a privacy researcher was able to use a publicly released medical data set to deduce William Weld, then Governor of Massachusetts, medical diagnoses and prescriptions (Barth-Jones 2012). Differential privacy marks a leap forward in the science of privacy by putting it on rigorous footing and away from past ad hoc and obscure notions of privacy. Several recent high profile implementations of differential privacy include Apple (Tang et al. 2017), Google (Erlingsson, Pihur, and Korolova 2014), Microsoft (Ding, Kulkarni, and Yekhanin 2017), and the U.S. Census Bureau (Abowd 2018).

Many data sets that are a target for deploying differential privacy contain valuable information that stake holders are still interested in learning about. However, the noise introduced by differential privacy changes the calculus of inference. As an example, we can implement differential privacy for tabular data by directly adding independent, random error to each cell; the amount and type of which is determined by theory. When we fit a regression model to the noise infused data, this will correspond to having measurement errors in the covariates. This, unfortunately, violates the assumptions of most statistical models. In the presence of such errors, standard estimators can exhibit significant bias and incorrect uncertainty quantification (Gong 2022; Karwa, Kifer, and Slavković 2015; Wang et al. 2018). These issues are a serious concern for researchers (Santos-Lozada, Howard, and Verdery 2020; Kenny et al. 2021; Winkler et al. 2021). Therefore, developing privacy-aware statistical workflows is necessary in order for science and privacy to coexist.

Unfortunately, making the necessary adjustments poses formidable mathematical challenges (Williams and Mcsherry 2010), even for seemingly simple models like linear regression. The difficulty lies in the marginal likelihood that results from correctly accounting for the injected privacy noise. This function is often analytically intractable and as a result, it is difficult or impossible to apply traditional statistical methods to derive estimators. In particular, the marginal likelihood can involve a complex integral where it is not even possible to evaluate the likelihood at a point. Tackling the problem by approximating the likelihood can be computationally unfeasible since the integral is usually high dimensional. Few tools are available to researchers to address these issues, and their absence is a serious barrier to the wider adoption of differential privacy.

The dapper package provides a set of tools for conducting privacy-aware Bayesian inference. It serves as a user-friendly R interface for the data augmentation framework proposed by Ju et al. (2022), allowing existing Bayesian models to be extended to handle noise infused data. The package is designed to integrate well with existing Bayesian workflows; results can be analyzed using tools from the rstan ecosystem in a drop-in fashion. Additionally, construction of a privacy-aware sampler is simplified through the specification of four independent modules. The benefits are twofold: several privacy mechanisms — these can even be from different formal privacy frameworks — can be compared easily by only swapping out relevant modules. And privacy mechanisms that have non-smooth transformations resulting in aforementioned intractable likelihoods (see example 3) can be incorporated with little work. As a result, dapper may prove particularly useful to those engaged in studying the privacy utility trade-off or dealing with a privacy mechanism that involves multiple transformations.

The rest of this article is organized as follows: Section 2 covers the necessary background to understand the mathematical notation and ideas used throughout the paper. Section 3 goes over the main algorithm without going into mathematical detail; for specifics see Ju et al. (2022). Section 4 provides an overview of the dapper package and discusses important implementation details. Section

5 contains two examples of how one might use the package to analyze the impact of adding noise for privacy. The first example goes over a typical odds ratio analysis for a 2×2 table and the second example covers a linear regression model.

2 Background

Let $x = (x_1, \dots, x_n) \in \mathcal{X}^n$ represent a confidential database containing n records. We will assume the data is generated by some statistical model $f(x | \theta)$. It is often the case some function of θ has relevant meaning to the scientific question at hand. In this setting, learning characteristics of a population reduces to learning about θ .

In the Bayesian statistical framework, learning about θ is accomplished by using the data x to update the posterior $p(\theta | x) \propto f(x | \theta)p(\theta)$. Here, $p(\theta)$ is called the prior distribution, and represents the researcher's belief about θ before seeing any data. The posterior represents uncertainty around θ and is formed by using Baye's rule to fuse together the observed data and the research's prior belief. One major advantage of the Bayesian method is that, through the prior, it provides a mechanism for incorporating information not explicitly contained in the data at hand. This is especially useful in settings where there is considerable domain knowledge on the value of θ .

For large data sets, it is common to work with a summary statistic $s = s(x)$ that has much smaller dimension than the original data. Doing so can greatly simplify calculations. In general, there can be information loss with using summary statistics, but for models with a sufficient statistic, there is no loss. Curators of large databases often use summary statistics to publish data since it allows them to efficiently communicate information contained in large data sets. For this reason, summary statistics are a natural target for dissemination based privacy approaches.

Differential Privacy

While a summary statistic can already partially anonymize data, it is still possible to deduce information about an individual entry depending on the distribution of x . Differential privacy (DP) offers a more principled approach by introducing randomness in a way that the output distribution does not change much when one individual's data is changed. A common approach – and the one dapper is primarily designed to address – is to take a summary statistic s , and add noise to it to produce a noisy summary statistic s_{dp} .

While noise addition has been common practice in statistical disclosure control (Dalenius and Reiss 1982), differential privacy provides a rigorous framework to specify where and how much noise to add. Most importantly, for the analyst, the specification of the differentially private noise mechanism can be made available without compromising privacy and incorporated into subsequent analyses.

The dapper package provides a very flexible framework that can accommodate the many different flavors a differential privacy; the main requirement being that the DP mechanism has a closed-form density. The section on privacy mechanisms provides details about utility functions for working with dapper in the concentrated differential privacy framework. However, for presentation, in this section we focus on the earliest and most common formulation of differential privacy, ϵ -differential privacy (ϵ -DP). The ϵ parameter is called the privacy loss budget. The privacy loss budget controls how strong the privacy guarantee is. Larger values of ϵ correspond to weaker privacy guarantees which in turn means less noise being added.

We now describe the ϵ -DP privacy framework in more detail. For the noisy summary statistic, we write $s_{dp} \sim \eta(\cdot | x)$. Here, η is a known noise infusion process designed to meet a certain property: The privacy mechanism η is said to be ϵ -differentially private (Dwork et al. 2006) if for all values of s_{dp} , and all “neighboring” databases $(x, x') \in \mathcal{X}^n \times \mathcal{X}^n$ differing by one record (denoted by $d(x, x') \leq 1$), the probability ratio is bounded:

$$\frac{\eta(s_{dp} | x)}{\eta(s_{dp} | x')} \leq \exp(\epsilon), \quad \epsilon > 0.$$

The differential privacy framework is used to create and verify privacy mechanisms. One such mechanism is the *Laplace mechanism*. It works by taking a deterministic statistic $s : X \mapsto \mathbb{R}^m$ and constructs the privatized statistic $s_{dp} := s(x) + u$ where u is a m -dimensional vector of i.i.d. Laplace random variables. The amount of noise, u , is scaled proportionally to the *global sensitivity* (or just sensitivity) of the statistic s . We define the sensitivity of a statistic s as $\Delta(s) := \max_{(x, x') \in \mathcal{X}^n \times \mathcal{X}^n, d(x, x') \leq 1} \|s(x) - s(x')\|$. If we draw each $u_i \sim \text{Lap}(\Delta(s)/\epsilon)$, we can show s_{dp} is ϵ -differentially private for the the Laplace mechanism. Example 2, will cover an application of the

Laplace mechanism to linear regression. Other common noise adding mechanisms include the Gaussian and the discrete Gaussian mechanisms, which also add noise scaled to the sensitivity; however these mechanisms satisfy a different privacy criteria called *zero concentrated differential privacy* (Bun and Steinke 2016).

3 Methodology

Given privatized data, s_{dp} , the goal of Bayesian inference is to sample from the posterior distribution $p(\theta \mid s_{dp})$. Since the observed likelihood, $p(s_{dp} \mid \theta)$, often has no simple closed form expression (Williams and Mcsherry 2010), most standard approaches do not apply. To conduct privacy-aware Bayesian inference, the dapper package implements the data augmentation algorithm of Ju et al. (2022) which allows us to sample from $p(\theta \mid s_{dp})$ without requiring a closed-form expression proportional to $p(s_{dp} \mid \theta)$. The algorithm accomplishes this by considering the joint distribution $p(\theta, x \mid s_{dp})$ and alternates sampling between the two distributions $p(\theta \mid x, s_{dp})$ and $p(x \mid \theta, s_{dp})$.

Since s_{dp} is derived from x , we have $p(\theta \mid x, s_{dp}) = p(\theta \mid x)$ which is just the usual posterior distribution given the confidential data x . The dapper package assumes the user has access to a sampler for $p(\theta \mid x)$. This can come from any R package such as `fmcmm` or constructed analytically via posterior conjugacy. For the second distribution, $p(x \mid \theta, s_{dp})$ may only be known up to a constant. The dapper package samples from this distribution by running a Gibbs-like sampler. Each of the n components of x is individually updated. However unlike the standard Gibbs sampler, each component is updated using a Metropolis-Hastings algorithm. This method is sometimes called the Metropolis-within-Gibbs sampler (Robert and Casella 2004).

In some cases, sampling from $p(x \mid \theta, s_{dp})$ can be made more efficient when the privacy mechanism can be written as a function of s_{dp} and a sum consisting of contributions from each individual record. More precisely, we say the privacy mechanism satisfies the *record additivity* property if

$$\eta(s_{dp} \mid x) = g\left(s_{dp}, \sum_{i=1}^n t_i(x_i, s_{dp})\right)$$

for some known and tractable functions $g, t_1, \dots, t_n$. The sample mean is an example of a summary statistic satisfying record additivity where $t_i(x_i, s_{dp}) = x_i$.

The data augmentation algorithm is described in the following pseudo code:

1. Sample θ^{t+1} from $p(\cdot \mid x^{(t)})$.
2. Sample from $p(x \mid \theta, s_{dp})$ using a three step process
 - Propose $x_i^* \sim f(\cdot \mid \theta)$.
 - If s satisfies the record additive property then update $s(x^*, s_{dp}) = t(x, s_{dp}) - t_i(x_i, s_{dp}) + t_i(x_i^*, s_{dp})$.
 - Accept the proposed state with probability $\alpha(x_i^* \mid x_i, x_{-i}, \theta)$ given by:

$$\alpha(x_i^* \mid x_i, x_{-i}, \theta) = \min\left\{1, \frac{\eta(s_{dp} \mid s(x_i^*, x_{-i}))}{\eta(s_{dp} \mid s(x_i, x_{-i}))}\right\} = \min\left\{1, \frac{g(s_{dp}, t(x^*, s_{dp}))}{g(s_{dp}, t(x, s_{dp}))}\right\}.$$

Theoretical results such as bounds on the acceptance probability as well as results on ergodicity can be found in Ju et al. (2022).

4 The Structure of dapper

The dapper package is structured around the two functions `dapper_sample()` and `new_privacy()`. The function, `dapper_sample()`, is used to generate MCMC draws from the private posterior. Targeting the correct private posterior requires a large set of inputs. In order to simplify the process of setting up the sampler, the `dapper_sample()` function uses a `privacy` object to encapsulate all information about the data generating process. The role of `new_privacy()` is to construct `privacy` objects. This separates inputs describing the data generating process from inputs describing simulation parameters, which decreases the chance for input related bugs.

Utility functions facilitating work with count data are also included. These center around the mass function and random number generators of the discrete Gaussian and discrete Laplacian distributions.

Privacy Model

Creating a privacy model is done using the `new_privacy()` constructor. The main arguments consist of the four components as outlined in the methodology section.

```
new_privacy(post_f = NULL, latent_f = NULL, priv_f = NULL,
            st_f = NULL, npar = NULL)
```

To minimize the potential for bugs, there are a set of requirements the four main components must adhere to which are described below:

- `latent_f()` is a function that samples from the parametric model describing how to generate a new confidential database x given model parameters θ . Its syntax must be `latent_f(theta)` where `theta` is a numeric vector representing the model parameters. This function must work with the `init_par` argument of `dapper_sample()`. The output must be a $n \times p$ numeric matrix where n is the number of observations and p is the dimension of a record x . The matrix requirement is strict, even if $p = 1$ `latent_f()` should return a $n \times 1$ matrix and not a vector of length n .
- `post_f()` is a function which makes a one-step draw from the private posterior. It has the syntax `post_f(dmat, theta)`. Here `dmat` is a numeric matrix representing the confidential database and `theta` is a numeric vector which serves as the initialization point for a one sample draw. The easiest, bug-free way to construct `post_f()` is to use a conjugate prior. However, this function can also be constructed by wrapping a MCMC sampler generated from other R packages (e.g. `rstan`, `fmcmm`, `adaptMCMC`). Using this approach requires caution; `dapper` requires a valid draw and many sampler implementations violate this requirement. This is especially true for adaptive samplers like `rstan`'s HMC where the first few draws are used to initialize the gradient and do not necessarily correspond to draws from a valid MCMC chain. Additionally, some packages like `mcmc` will generate samplers that may be slow due to a large initialization overhead. For these reasons we recommend sticking with conjugate priors as they will be quick and avoid serious undetected semantic errors arising from specific implementation details of other R packages.
- `priv_f()` is a function that returns the log of the privacy mechanism density given the noise-infused summary statistics s_{dp} and its potential true value $s(x, s_{dp}) := \sum_{i=1}^n t_i(x_i, s_{dp})$. This function has the syntax `priv_f(sdp, sx)` where `sdp` and `sx` are numeric vectors or matrices representing the value of s_{dp} and $s(x, s_{dp})$ respectively. The arguments must appear in the exact order with the same variable names as defined above. Finally, the return value of `priv_f()` must be a numeric vector of length one.
- `st_f()` is a function which calculates a summary statistic. It must be defined using the three arguments named `i`, `xi` and `sdp` in the stated order. The role of this function is to represent terms in the definition of record additivity with each of the three arguments in `st_f` corresponding to the similarly spelled terms in $t_i(x_i, s_{dp})$. Here `i` is an integer, while `xi` is a numeric vector and `sdp` is a numeric vector or matrix. The return value must be a numeric vector or matrix.
- `npar` is an integer equal to the dimension of θ .

Sampling

The `dapper_sample()` function essentially takes an existing Bayesian model and extends it to handle privatized data. The output of `dapper_sample()` contains MCMC draws from the private posterior. The function has syntax:

```
dapper_sample(data_model, sdp, init_par, niter = 2000, warmup = floor(niter / 2),
              chains = 1, varnames = NULL)
```

The parameters `data_model`, `sdp`, and `init_par` are required. The `data_model` input is a privacy model object that is constructed using `new_privacy()` (see section [Privacy Model](#)). The value of `sdp` is equal to the observed noise infused statistic. We require the object class of `sdp` to be the same as the output of `st_f()`. For example, if `st_f()` returns a matrix then `sdp` must also be a matrix. The provided starting value of the chain (`init_par`) must work with the `latent_f()` component. An error will be thrown if `latent_f()` evaluated at `init_par` does not return a numeric matrix.

The optional arguments are the number of MCMC draws (`niter`), the burn in period (`warmup`), number of chains (`chains`) and character vector that names the parameters. Running multiple chains can be done in parallel using the `furrr` package. Additionally, progress can be monitored using the

`progressr` package. Adhering to the design philosophy of the two packages, we leave the setup to the user so that they may choose the most appropriate configuration for their system. The contingency table demonstration given in section 5 walks through a typical setup of `furrr` and `progressr`.

The return value of `dapper_sample()` is a list containing a `draw_matrix` object and a vector of acceptance probabilities of size `niter`. The `draw_matrix` object is described in more detail in the `posterior` package. The advantages with working with a `draw_matrix` object is that it is compatible with many of the packages in the `rstan` ecosystem. For example, any `draw_matrix` object can be plugged directly into the popular `bayesplot` package. Additionally, `dapper`'s basic summary function provides the same posterior summary statistics as those found when using `rstan`. Overall, this should make working with `dapper` easier for anyone already familiar with the `rstan` ecosystem.

Privacy Mechanisms for Count Data

The `dapper` package provides several utility functions for analyzing privatized count data. Currently, the U.S. Census is a major driver behind the deployment and research of privatized count data, and these functions were created for census oriented researchers in mind.

Pure ϵ -differential privacy is a stringent requirement that can lead to poor data utility. For this reason the U.S. Census uses zero concentrated differential privacy which is a weaker criteria. The `dapper` package implements probability mass and sampling functions for the discrete Gaussian and discrete Laplace distributions (Canonne, Kamath, and Steinke 2021). These are two potential mechanisms for deploying concentrated differential privacy, and both are, in fact, used in several current US Census data products (Labs 2022).

The discrete Gaussian / Laplacian mechanism have probability mass functions given in the panel below.

$$P[X = x] = \frac{e^{-(x-\mu)^2/2\sigma^2}}{\sum_{y \in \mathbb{Z}} e^{-(y-\mu)^2/2\sigma^2}} \quad (\text{Discrete Gaussian})$$

$$P[X = x] = \frac{e^{1/t} - 1}{e^{1/t} + 1} e^{-|x|/t} \quad (\text{Discrete Laplacian})$$

For both distributions $x \in \mathbb{Z}$. The discrete Gaussian has two parameters (μ, σ) which govern the location and scale respectively. The functions `ddnorm` and `rdnorm` provide the density and sampling features for the discrete Gaussian distribution. The `ddnorm` function contains a calculation for the normalizing constant which is expensive. To speed up repeated execution, memoization is used via the `memoise` package.¹ Finally, the functions `ddlplace` and `rdlplace` provide similar features for the discrete Laplace distribution. The implementations of these functions are not the safest and are meant to provide uses with a way to quickly explore the behavior of DP mechanisms.

5 Examples

Example 1: 2x2 Contingency Table (Randomized Response)

As a demonstration, we analyze a subset of the UC Berkeley admissions data, which is often used as an illustrative example of Simpson's paradox (Bickel, Hammel, and O'Connell 1975). The question posed is whether the data suggest there is bias against females during the college admissions process. Below is a table of the aggregate admissions result from six departments based on sex for a total of $N = 400$ applicants. The table on the left represents the true admissions data and the table on the right is the result of anonymizing the data with a differentially private mechanism.

	Admitted	Rejected		Admitted	Rejected
Female	46	118	Female	74	102
Male	109	127	Male	104	120

To see how the privacy mechanism works, we envision the record level data set as a $N \times 2$ matrix with the first column representing sex and the second column representing admission status. Thus each row in the matrix is the response of an individual. To anonymize the results, we apply a random response scheme where for each answer we flip a fair coin twice.² As a concrete example, suppose

¹Memoization is a programming technique used to cache repeated computations

²The randomized response scheme predates the development of differential privacy and was first described by Warner (1965) as a means to reduce survey bias involving sensitive questions.

Robert is a male who was rejected. To anonymize his response, we would first flip a coin to determine if his sex response is randomized. If the first flip is heads we keep his original response of being a male. If we see tails, then we would flip the coin again and change the answer to male or female depending on whether we see heads or tails respectively. We then repeat this process for his admission status. This anonymization scheme conforms to a mechanism with a privacy budget of at most $\epsilon = 2 \log(3)$.

To set up dapper to analyze the anonymized admissions data, we first encode our anonymized record level data using a binary matrix where male and admit take the value 1. From this we can construct s_{dp} as the columns of the binary matrix stacked on top of each other.

1. `latent_f`: For each individual there are four possible sex/status responses which can be modeled using a multinomial distribution. To implement draws from the multinomial distribution we use the `sample` function to take samples from a list of containing the four possible binary vectors. Note the final line results in a 400×2 matrix.

```
latent_f <- function(theta) {
  t1 <- list(c(1,1), c(1,0), c(0,1), c(0,0))
  rs <- sample(t1, 400, replace = TRUE, prob = theta)
  do.call(rbind, rs)
}
```

2. `post_f`: Given confidential data, we can derive the posterior analytically using a Dirichlet prior. In this example, we use a flat prior which corresponds to Dirichlet(1) distribution. A sample from the Dirichlet distribution can be generated using random draws from the gamma distribution.

```
post_f <- function(dmat, theta) {
  sex <- dmat[,1]
  status <- dmat[,2]

  #Male & Admit
  x1 <- sum(sex & status)
  x2 <- sum(sex & !status)
  x3 <- sum(!sex & status)
  x4 <- sum(!sex & !status)

  x <- c(x1, x2, x3, x4)

  t1 <- rgamma(4, x + 1, 1)
  t1/sum(t1)
}
```

3. `st_f`: The private summary statistic s_{dp} can be written as a record additive statistic using indicator functions. Let v_i be a binary vector of length 2×400 where the entries with index i and $400 + i$ are the only possible non zero entries. We let these two entries correspond to the sex and admission status response of the individual with record x_i . With this construction we have $s_{dp} = \sum_{i=1}^{400} v_i$.

```
st_f <- function(xi, sdp, i) {
  x <- matrix(0, nrow = 400, ncol = 2)
  x[i,] <- xi
  x
}
```

4. `priv_f`: The privacy mechanism is the result of two fair coin flips, so for each answer there is a $3/4$ chance it remains the same and a $1/4$ chance it changes. Hence the log likelihood of observing s_{dp} given the current value of the latent database, tx , is $\log(3/4)$ times the number of entries that match plus $\log(1/4)$ times the number of entries which differ.

```
priv_f <- function(sdp, sx) {
  t1 <- sum(sdp == sx)
  t1 * log(3/4) + (800 - t1) * log(1/4)
}
```

Below we load the data and create the noisy admissions table.

```
#Original UCBAmissions data.
cnf_df <- tibble(sex = c(1, 1, 0, 0),
                 status = c(1, 0, 1, 0),
                 n = c(1198, 1493, 557, 1278)) %>% uncount(n)

set.seed(1)
ix <- sample(1:nrow(cnf_df), 400, replace = FALSE)
cnf_df <- cnf_df[ix,]

#Answers to be randomized
ri <- as.logical(rbinom(800, 1, 1/2))

#Randomized answers
ra <- rbinom(sum(ri), 1, 1/2)

#Create sdp
sdp <- as.matrix(cnf_df)
sdp[ri] <- ra
```

Once we have defined all components of the model we can create a new privacy model object using the `new_privacy` function and feed this into the `dapper_sample` function. Below we run four chains in parallel each with 5,000 posterior draws with a burn-in of 1000.

```
library(dapper)
library(furrr)
plan(multisession, workers = 4)

dmod <- new_privacy(post_f = post_f,
                    latent_f = latent_f,
                    priv_f = priv_f,
                    st_f = st_f,
                    npar = 4,
                    varnames = c("pi_11", "pi_10", "pi_01", "pi_00"))

dp_out <- dapper_sample(dmod,
                       sdp = sdp,
                       seed = 123,
                       niter = 6000,
                       warmup = 1000,
                       chains = 4,
                       init_par = rep(.25,4))
```

If the run time of `dapper_sample` is exceptionally long, one can use the `progressr` package to monitor progress. The `progressr` framework allows for a unified handling of progress bars in both the sequential and parallel computing case.

```
library(progressr)
handlers(global = TRUE)

handlers("cli")
dp_out <- dapper_sample(dmod,
                       sdp = sdp,
                       seed = 123,
                       niter = 6000,
                       warmup = 1000,
                       chains = 4,
                       init_par = rep(.25,4))
```

Results can be quickly summarized using the `summary` function which is displayed below. The `rhat` values in the table are close to 1, which indicates the chain has run long enough to achieve adequate mixing.

```
#> # A tibble: 4 x 10
#>   variable mean median    sd    mad    q5   q95  rhat ess_bulk ess_tail
```

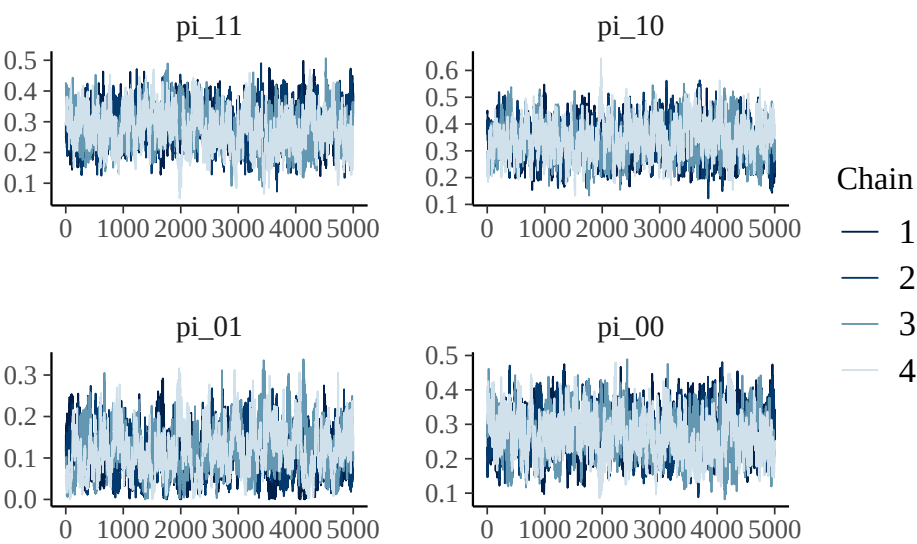


Figure 1: (Example 1) trace plots.

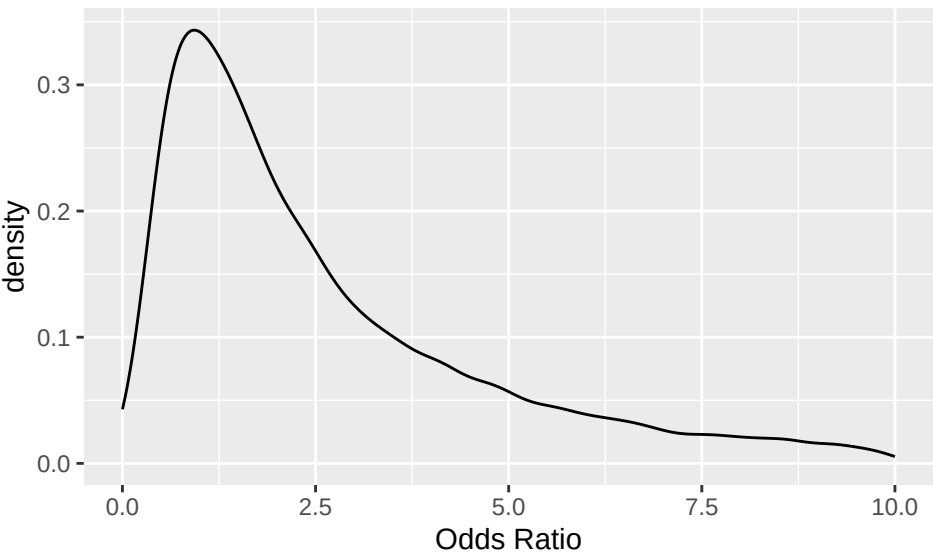


Figure 2: (Example 1) posterior density estimate for the odds ratio using 16,000 MCMC draws.

```
#>   <chr>   <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>   <dbl> <dbl>
#> 1 pi_11  0.281 0.281 0.0610 0.0625 0.182 0.382 1.02    362.    818.
#> 2 pi_10  0.336 0.335 0.0638 0.0640 0.235 0.444 1.01    431.   1191.
#> 3 pi_01  0.111 0.108 0.0548 0.0563 0.0250 0.206 1.02    282.    504.
#> 4 pi_00  0.272 0.272 0.0601 0.0616 0.172 0.372 1.02    389.    875.
```

Diagnostic checks using trace plots can be done using the [Bayesplot](#) package as shown in figure 1. It is especially important to check for good mixing with dapper since sticky chains are likely to be produced when the amount of injected noise is high. See the discussion section for a more detailed explanation.

To see if there is evidence of gender bias we can look at the odds ratio. Specifically, we look at the odds of a male being admitted to that of female. A higher odds ratio would indicate a bias favoring males. Figure 2 shows draws from the private posterior. The large odds ratio values would seem to indicate there is bias favoring the males. The “paradox” arises when analyzing the data stratified by university department where the odds ratio flips with females being favored over males.

For comparison, we run a standard Bayesian analysis on the noise infused table ignoring the privacy mechanism. This will correspond exactly to the model defined in the `post_f` component. Figure 3 shows a density estimate for the odds ratio under the confidential and noisy data. The

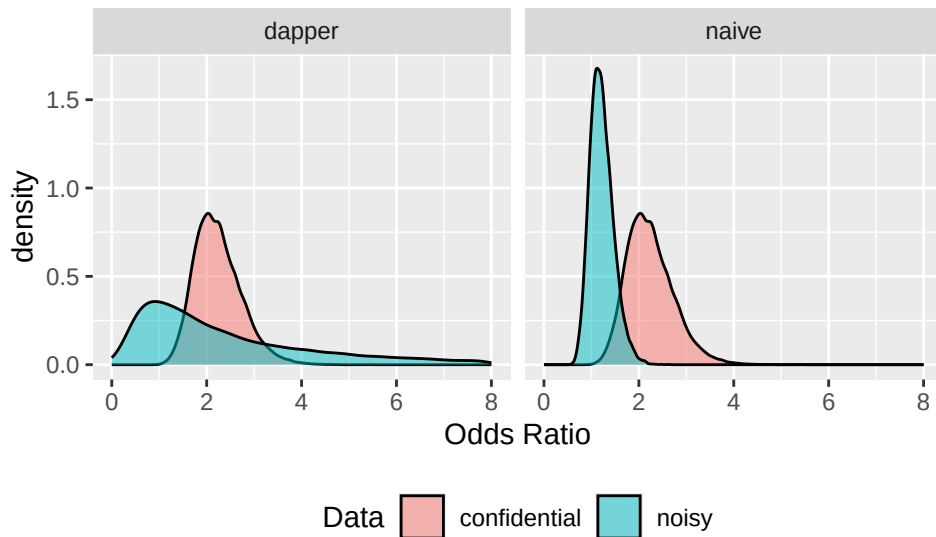


Figure 3: (Example 1) Posterior distributions (cyan) of the odds ratio (admission of males vs. females) using noisy (i.e. privacy-protected) data. Left panel: correct Bayesian inference using dapper which takes into account the privacy mechanism; Right panel: naive Bayesian inference treating the noisy data as noise-free. Red distribution in both panels reflect the true posterior distribution if the analysis were to be conducted on the confidential data.

posterior distribution for the odds ratio under the noisy data is shifted significantly, indicating a large degree of bias. Looking at left hand plot in figure 3 shows the MAP estimate from dapper is similar to that in the case of the confidential data. The width of the posterior is also much larger since it properly accounts for the uncertainty due to the privacy mechanism. This illustrates the dangers of ignoring the privacy mechanism: a naive analysis not only has bias, but also severely underestimates the uncertainty associated with the odds ratio estimate.

Example 2: 2x2 Contingency Table (Discrete Gaussian)

To highlight the flexibility of dapper we reanalyze example 1 with a different privacy mechanism. Here we take inspiration from the 2020 U.S. Decennial Census, which is in the process of deploying a novel privacy protection system for count data. Because there are many users of Census data, a successful deployment requires a careful analysis of the privacy / utility trade-off and includes studying how different mechanisms and privacy parameters affect down stream analyses.

One of the mechanisms used is the discrete Gaussian distribution. In our admissions data example, this privacy mechanism works by injecting noise into the the total cell counts given in the 2x2 table. The randomized response scheme, in contrast, injects noise at the record level. The dapper package accommodates both the randomized response and the discrete Gaussian mechanisms, allowing us to compare the impact of the two approaches.

To begin comparing the two approaches, we need to set the privacy parameters. Since the two frameworks use different metrics, there does not exist a direct comparison. However, there is a direct relationship between zCDP and (ϵ, δ) -differential privacy. The latter framework is a relaxed version of ϵ -DP where the ratio bound only holds in probability. To make our comparison we will set $\delta = 10^{-10}$, which is the value the U.S. Census uses. To make a comparison, we set the scale parameter in the discrete Gaussian distribution to 6.25 which will guarantee $(2 \log(3), 10^{-10})$ -differential privacy.³

The table below on the right represents a hypothetical data protected using a discrete Gaussian mechanism.

For the public, the US Census only releases aggregate cell counts along with the true total count of said cells. In our example, this means the Census would have released the right hand table along with the fact that $N = 400$ in the original table. Thus, it natural to let s_{dp} be the vector of cell counts. As in example 1, we imagine the latent database as a 400×2 binary matrix. Below we describe the process

³ $\frac{1}{2}\epsilon^2$ -concentrated differential privacy implies $\left(\frac{1}{2}\epsilon^2 + \epsilon\sqrt{2\log(1/\delta)}, \delta\right)$ -differential privacy (Bun and Steinke 2016)

	Admitted	Rejected		Admitted	Rejected
Female	46	118	Female	47	110
Male	109	127	Male	110	131

for analyzing the privatized data using dapper. Since the latent process and posterior are the same as example 1, we only describe how to construct `st_f` and `priv_f`.

1. `st_f`: The private summary statistic s_{dp} can be written as a record additive statistic using the indicator vectors $(1, 0, 0, 0)$, $(0, 1, 0, 0)$, $(0, 0, 0, 1)$ and $(0, 0, 0, 1)$. These four vectors correspond to the four possible cells.

```
st_f <- function(xi, sdp, i) {
  if(xi[1] & xi[2]) {
    c(1, 0, 0, 0)
  } else if (xi[1] & !xi[2]) {
    c(0, 1, 0, 0)
  } else if (!xi[1] & xi[2]) {
    c(0, 0, 1, 0)
  } else {
    c(0, 0, 0, 1)
  }
}
```

2. `priv_f`: The privacy mechanism us a discrete Gaussian distribution centered at 0.

```
priv_f <- function(sdp, sx) {
  sum(dapper::ddnorm(sdp - sx, mu = 0, sigma = 6.25, log = TRUE))
}
```

The summary table below show the results of running a chain for 2000 iterations with a burn-in of 1000 runs.

```
summary(dp_out)
```

```
#> # A tibble: 4 x 10
#>   variable mean median    sd    mad    q5    q95  rhat ess_bulk ess_tail
#>   <chr>    <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>    <dbl>    <dbl>
#> 1 pi_11    0.315   0.313  0.0266 0.0269 0.273  0.360  1.00    660.    684.
#> 2 pi_01    0.286   0.285  0.0267 0.0269 0.243  0.329  1.00    611.    622.
#> 3 pi_10    0.107   0.106  0.0197 0.0200 0.0767 0.141  1.00    428.    737.
#> 4 pi_00    0.292   0.292  0.0262 0.0261 0.251  0.339  1.00    666.    867.
```

Figure 5 juxtaposes the private posterior under the randomized(dashed line) and discrete Gaussian (solid line) mechanism. Comparing the two suggest using discrete Gaussian noise leads to slightly less posterior uncertainty and a mode closer to the true, confidential posterior.

Additionally, if we compare the left hand plots of figure 3 and figure 5, the discrete Gaussian induces considerably less bias when using the naive analysis.

Example 3: Linear Regression

In this section we apply dapper to reconstruct an example presented in Ju et al. (2022). They apply a Laplace privacy mechanism to a sufficient summary statistic for a linear regression model. Let $\{(x_i, y_i)\}_{i=1}^n$ be the original, confidential data with $x_i \in \mathbb{R}^2$. They assume the true data generating process follows the model

$$\begin{aligned}
 y &= -1.79 - 2.89x_1 - 0.66x_2 + \epsilon \\
 \epsilon &\sim N(0, 2^2) \\
 \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} &\sim N_2(\mu, I_2) \\
 \mu &= \begin{pmatrix} 0.9 \\ -1.17 \end{pmatrix}.
 \end{aligned}$$

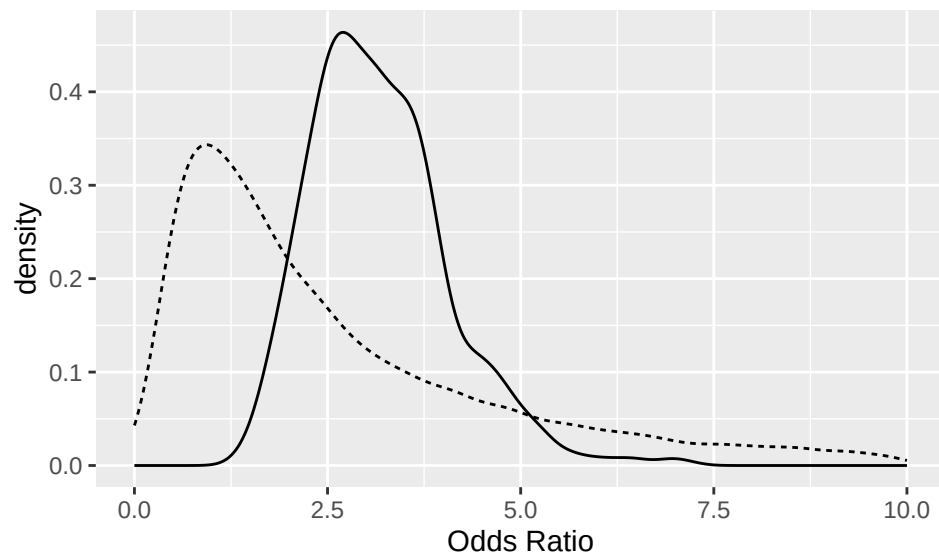


Figure 4: (Example 2) posterior density estimate for the odds ratio using 1,000 MCMC draws.

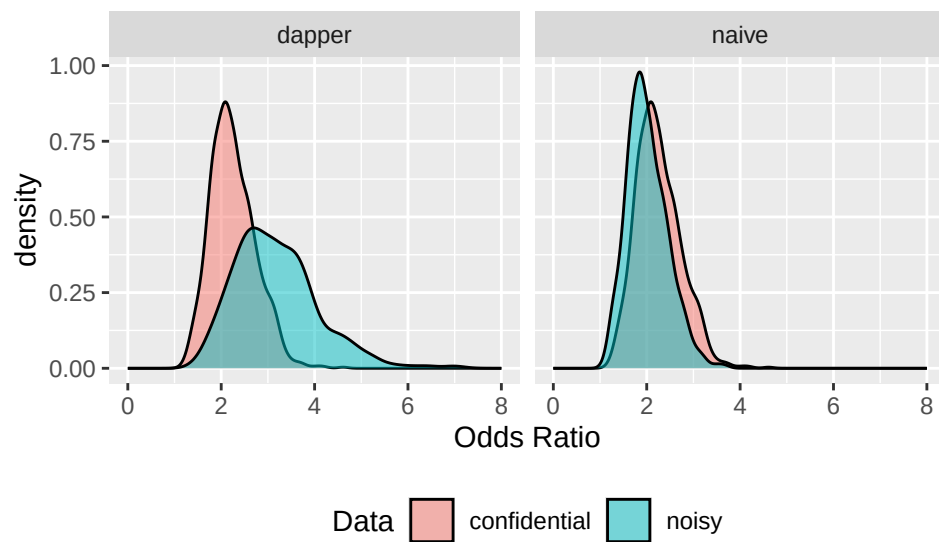


Figure 5: (Example 2) comparison between using dapper and a naive Bayesian analysis on the noise infused data and the original confidential data.

Note, in most settings involving linear regression, the covariates are assumed to be fixed, known constants. Thus the formulation above is a departure from the norm since we are assuming a random design matrix. More details on why this framing is necessary will be provided later when describing the latent model. The paper considers the scenario where one desires to publicly release the sufficient summary statistics

$$s(x, y) = (x^T y, y^T y, x^T x).$$

This summary statistic satisfies the additive record property since $s(x, y) = \sum_{i=1}^n t(x_i, y_i)$ where

$$t(x_i, y_i) = ((x_i)^T y_i, y_i^2, (x_i)^T x_i),$$

with ϵ -DP privacy guarantees. To achieve this guarantee it is necessary to bound the value of the statistic. This will ensure a data point cannot be too “unique.” Intuitively, an outlier is easy to identify, so we want to make sure no data point is an extreme outlier. In the language of differential privacy, this equates to bounding the global sensitivity. This was accomplished by clamping the data. More precisely, we define the clamp function $[z] := \min\{\max\{z, -10\}, 10\}$ which truncates a value z so that it falls into the interval $[-10, 10]$. Furthermore, we let $\tilde{z} := [z]/10$ denote the normalized clamped value of z . The clamped statistic is

$$t(x_i, y_i) = ((\tilde{x}^i)^T \tilde{y}_i, \tilde{y}_i^2, (\tilde{x}_i)^T \tilde{x}_i).$$

Ignoring duplicate entries, the global sensitivity is $\Delta = p^2 + 4p + 3$.⁴ Using the Laplace mechanism, ϵ -DP privacy can thus be achieved by adding i.i.d. $\text{Laplace}(0, \Delta/\epsilon)$ error to each unique entry. A tighter bound on sensitivity can be achieved using other techniques, see Awan and Slavković (2020).

1. `latent_f`: Since the privacy mechanism involves injecting noise into the design matrix, it is not possible to use the standard approach where one assumes the design matrix is a fixed, known constant. Hence to draw a sample from the latent data generating process we use the relation $f(x, y) = f(x)f(y | x)$. In this formulation, it is necessary to specify a distribution on the covariates x .

```
latent_f <- function(theta) {
  xmat <- MASS::mvrnorm(50, mu = c(.9, -1.17), Sigma = diag(2))
  y <- cbind(1, xmat) %*% theta + rnorm(50, sd = sqrt(2))
  cbind(y, xmat)
}
```

2. `post_f`: Given confidential data X we can derive the posterior analytically using a normal prior on β .

$$\begin{aligned}\beta &\sim N_{p+1}(0, \tau^2 I_{p+1}) \\ \beta | x, y &\sim N(\mu_n, \Sigma_n) \\ \Sigma_n &= (x^T x / \sigma^2 + I_{p+1} / \tau^2)^{-1} \\ \mu_n &= \Sigma_n (x^T y) / \sigma^2\end{aligned}$$

In the example, we use $\sigma^2 = 2$ and $\tau^2 = 4$.

```
post_f <- function(dmat, theta) {
  x <- cbind(1, dmat[, -1])
  y <- dmat[, 1]

  ps_s2 <- solve((1/2) * t(x) %*% x + (1/4) * diag(3))
  ps_m <- ps_s2 %*% (t(x) %*% y) * (1/2)

  MASS::mvrnorm(1, mu = ps_m, Sigma = ps_s2)
}
```

3. `st_f`: The summary statistic contains duplicate entries. We can considerably reduce the dimension of the statistic by only considering unique entries. The `clamp_data` function is used to bound the statistic to give a finite global sensitivity.

```
clamp_data <- function(dmat) {
  pmin(pmax(dmat, -10), 10) / 10
}
```

⁴The original paper, Ju et al. (2022), contains a computation error and mistakenly uses $\Delta = p^2 + 3p + 3$

```

st_f <- function(i, tx, sdp) {
  txc <- clamp_data(tx)
  ydp <- txc[1]
  xdp <- cbind(1,t(txc[-1]))

  s1 <- t(xdp) %*% ydp
  s2 <- t(ydp) %*% ydp
  s3 <- t(xdp) %*% xdp

  ur_s1 <- c(s1)
  ur_s2 <- c(s2)
  ur_s3 <- s3[upper.tri(s3,diag = TRUE)][-1]
  c(ur_s1,ur_s2,ur_s3)
}

```

4. `priv_f`: Privacy Mechanism adds $\text{Laplace}(0, \Delta/\epsilon)$ error to each unique entry of the statistic. In this example, $\Delta = 15$ and $\epsilon = 10$.

```

priv_f <- function(sdp, zt) {
  sum(VGAM::dlaplace(sdp - zt, 0, 15/10, log = TRUE))
}

```

First we simulate fake data using the aforementioned privacy mechanism. In the example, we use $n = 50$ observations.

```

deltaa <- 15
epsilon <- 10
n <- 50

set.seed(1)
xmat <- MASS::mvrnorm(n, mu = c(.9,-1.17), Sigma = diag(2))
beta <- c(-1.79, -2.89, -0.66)
y <- cbind(1,xmat) %*% beta + rnorm(n, sd = sqrt(2))

#clamp the confidential data in xmat
dmat <- cbind(y,xmat)
sdp <- apply(sapply(1:nrow(dmat), function(i) st_f(i, dmat[i,], sdp))), 1, sum)

#add Laplace noise
sdp <- sdp + VGAM::rlaplace(length(sdp), location = 0, scale = deltaa/epsilon)

```

We construct a privacy model using the `new_privacy` function and make 25,000 MCMC draws with a burn in of 1000 draws.

```

library(dapper)

dmod <- new_privacy(post_f = post_f,
                    latent_f = latent_f,
                    priv_f = priv_f,
                    st_f = st_f,
                    npar = 3,
                    varnames = c("beta0", "beta1", "beta2"))

dp_out <- dapper_sample(dmod,
                        sdp = sdp,
                        niter = 25000,
                        warmup = 1000,
                        chains = 1,
                        init_par = rep(0,3))

```

The output of the MCM run is reported below.

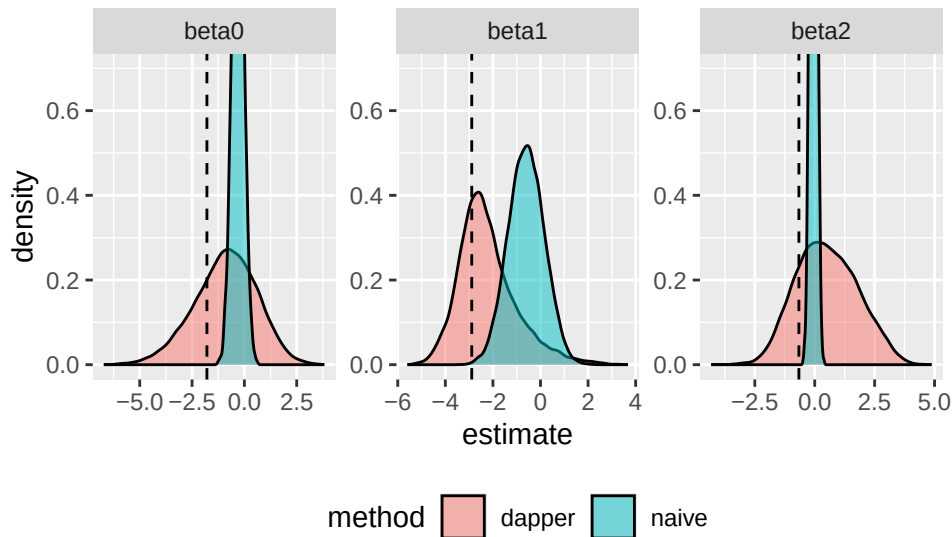


Figure 6: (Example 3) comparison between dapper and a naive approach that ignores the privacy mechanism. The dashed lines are the true coefficient values.

```
summary(dp_out)
```

```
#> # A tibble: 3 x 10
#>   variable mean median    sd mad    q5    q95  rhat ess_bulk ess_tail
#>   <chr>      <dbl>  <dbl> <dbl> <dbl> <dbl> <dbl>  <dbl>    <dbl>
#> 1 beta0    -0.900 -0.821  1.49  1.47 -3.49  1.41  1.00    471.    1082.
#> 2 beta1    -2.22 -2.42  1.22  1.03 -3.91  0.136 1.00    231.     329.
#> 3 beta2     0.511  0.454  1.28  1.36 -1.48  2.70  1.00    224.     480.
```

For comparison, we consider a Bayesian analysis where the design matrix is a fixed known constant and σ^2 is known. Using the diffuse prior $f(\beta) \propto 1$ leads to normal posterior.

$$f(\beta | x, y, \sigma^2) \sim N(\hat{\beta}, \hat{\Sigma})$$

$$\hat{\mu} = (x^T x)^{-1} x y$$

$$\hat{\Sigma} = \sigma^2 (x^T x)^{-1}$$

The posterior can be written as a function of $s(x, y)$. Since we only have access to the noisy version s_{dp} we can attempt to reconstruct the posterior by extracting the relevant entries which is done below.

Because of the injected privacy noise, the reconstructed $(x^T x)^{-1}$ matrix is not positive definite. As a naive solution we use the algorithm proposed by Higham (1988) to find the closest positive semi-definite matrix as determined by the Frobenius norm. The `pracma` package contains an implementation via the `nearest_psd` function.

Figure 6 shows the posterior density estimates for the β coefficients based on s_{dp} . The density estimates indicate the naive method, which ignores the privacy mechanism, has bias and underestimates the variance. Likewise Figure 7 illustrates how dapper provides point estimates that are not far off from those that would have been obtained using the original confidential data. And the dramatic increase in the posterior variance indicates the privacy mechanism adds substantial uncertainty to the estimates.

6 Discussion

Mixing and Privacy Loss Budget

Mixing can be poor when the posterior under a given privacy mechanism is much wider than the posterior that would arise using the confidential data. In other words, a small privacy budget can result in poor mixing. Intuitively, this issue arises because the step size of the chain is governed by the variance of the posterior model (step 1 of the algorithm) that assumes no privacy noise. Thus, a

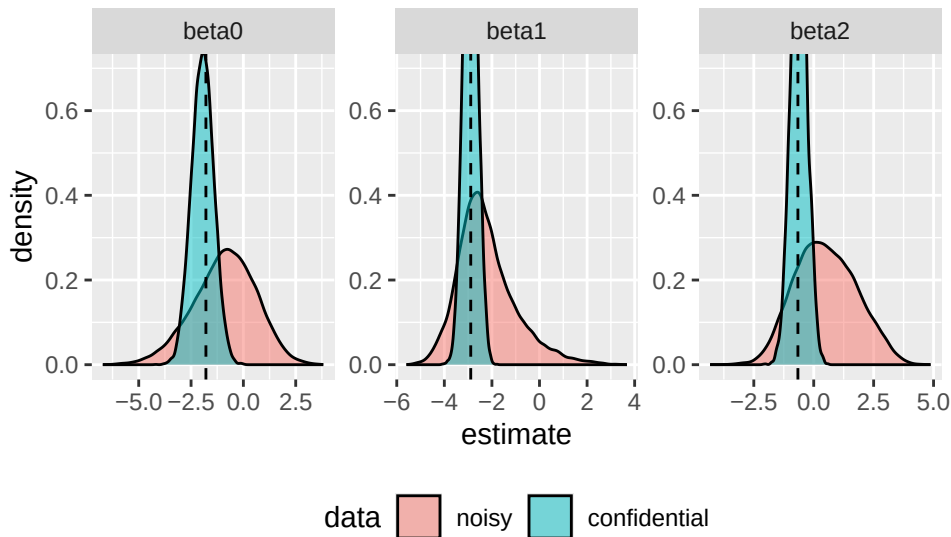


Figure 7: (Example 3) comparison for example between using dapper on the noisy data set and a standard Bayesian analysis on the confidential data set.

small privacy budget will generate a chain whose step sizes are too small to effectively explore the private posterior. The rest of this section explores a toy example that will provide insight into this phenomenon.

Suppose the confidential data consist of a single observation $x \in \mathbb{R}$, and consider the scenario where a user makes a request to view x and in return receives $s := x + \nu$, which is a noise infused version of x . For simplicity, we do not worry about constructing an ϵ -DP privacy mechanism, and take $\nu \sim N(0, \epsilon^{-2})$ for some $\epsilon > 0$. However, it will still be useful to think of ϵ as the privacy budget since smaller values of ϵ correspond to a larger amounts of noise. Using a flat prior and a normally distributed likelihood results in a normally distributed posterior described below.

$$\begin{aligned} f(\theta) &\propto 1 \\ s \mid x &\sim N(x, \epsilon^{-2}) \\ x \mid \theta &\sim N(\theta, \sigma^2) \end{aligned}$$

With the above model, the data augmentation process consist of the two steps

- Step 1: Sample from $x \mid \theta, s \sim N(\mu, \tau^2)$, where μ and τ are defined as:

$$\begin{aligned} \mu &:= \frac{\frac{1}{\epsilon^{-2}}s + \frac{1}{\sigma^2}\theta}{\frac{1}{\epsilon^{-2}} + \frac{1}{\sigma^2}} \\ \tau^2 &:= \frac{1}{\frac{1}{\epsilon^{-2}} + \frac{1}{\sigma^2}}. \end{aligned}$$

- Step 2: Sample from $\theta \mid x, s \sim N(x, \sigma^2)$.

In the setting of this example, Liu and Wu (1999) showed the Bayesian fraction of missing information gives the exact convergence rate. The Bayesian fraction of missing information, γ is defined as

$$\gamma := 1 - \frac{E[\text{Var}(\theta \mid s, x) \mid s]}{\text{Var}(\theta \mid s)} = 1 - \frac{E[\text{Var}(\theta \mid x)]}{\text{Var}(\theta \mid s)}.$$

Plugging in the appropriate quantities into the above panel gives us

$$\gamma = 1 - \frac{\sigma^2}{\sigma^2 + \epsilon^{-2}} = 1 - \frac{1}{1 + \frac{\epsilon^{-2}}{\sigma^2}}.$$

The chain converges faster as $\gamma \rightarrow 0$ and slower as $\gamma \rightarrow 1$. From the right hand term in the above panel, we can see γ depends only on ϵ^{-2}/σ^2 and as the privacy budget decreases (i.e. more noise is being added to x), $\gamma \rightarrow 1$.

Thus we recommend playing with the privacy budget to troubleshoot the cause of slow mixing chains. In particular it may be useful to use a relatively large privacy budget for qualitative comparisons so that the privacy / utility trade-off can be explored more easily. If a faster sampler is needed, and it has been determined that the privacy budget is the issue, the pseudo-likelihood scheme proposed by Andrieu and Roberts (2009) may offer significant speed ups. This scheme fits in the same data augmentation framework as dapper but is not implemented.

7 Summary

Currently, there is a lack of software tools privacy researchers can use to evaluate the impact of privacy mechanisms on statistical analyses. While there have been tremendous gains in the theoretical aspects of privacy, the lack of software resources to deploy and work with new privacy techniques has hampered their adoption. This gap in capability has been noted by several large industry entities who have begun building software ecosystems for working with differential privacy. However, the majority of these software tools only address privacy and not the ensuing analysis or, if it does, addresses the analysis only for specific models. Privacy researchers currently lack good tools for evaluating the impact of privacy mechanisms on a statistical analysis.

Thus dapper helps fill an urgent need by providing researchers a way to properly account for the noise introduced for privacy protection in their statistical analysis. A notable feature is its flexibility which allows the users to specify a custom privacy mechanism. The benefit being that dapper can evaluate already established privacy mechanisms as well as those that have yet to be discovered.

This package offers a significant step forward in providing general-purpose statistical inference tools for privatized data. Despite the strengths of dapper, it has several stringent requirements for good performance that limit its potential: 1) the privacy mechanism must have a closed-form density 2) a record additive statistic must be used to leverage dapper's full computational potential 3) the non-private posterior sampler needs to be efficient and 4) the privacy budget cannot be too small. To improve dapper, future work could aim to relax some of these requirements.

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